

Lecture 2 – Option Pricing Theory

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Lecture 2 – Option Pricing Theory

- Itô's Lemma
- Derivation of the Black–Scholes PDE via a hedging portfolio
- Martingales and Option Pricing
- Feynman–Kac Formula
- Analytical solution for the Black–Scholes PDE

Itô's lemma (1/3)

- Suppose X_t follows a diffusion process:

$$dX_t = \mu_t dt + \sigma_t dW_t, \quad \text{with } X_0 = x_0,$$

or

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s,$$

with μ_t being the drift term and σ_t being the diffusion term.

- Let now $g(t, X_t)$ be a function with continuous derivatives (up to the second order). Then $Y_t := g(t, X_t)$ follows the process:

$$dY_t = \left(\frac{\partial g}{\partial t} + \mu_t \frac{\partial g}{\partial X_t} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 g}{\partial X_t^2} \right) dt + \sigma_t \frac{\partial g}{\partial X_t} dW_t.$$

- Another useful stochastic calculus is Itô's product rule, i.e., for two diffusion processes Y_t and Z_t , it holds that

$$d(Y_t Z_t) = Z_t dY_t + Y_t dZ_t + dY_t dZ_t.$$

Itô's lemma (2/3)

- Itô's lemma can be intuitively understood using the formal rule

$$dW_t dW_t = dt.$$

- The increment of Brownian motion over $[t, t + dt]$ satisfies $W_{t+dt} - W_t \sim \mathcal{N}(0, dt)$, hence dW_t is of order $\sqrt{dt}$.
 - Therefore dW_t^2 is of order dt , while dt^2 and $dW_t dt$ are of higher order and vanish faster.
 - Since dW_t^2 is of order dt , it must be retained in stochastic calculus.
- An informal Taylor expansion of $g(t, X_t)$ is then

$$dg = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial X_t} dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial t^2} dt dt + \frac{1}{2} \frac{\partial^2 g}{\partial X_t^2} dX_t dX_t + \frac{\partial^2 g}{\partial X_t \partial t} dX_t dt + \dots$$

Itô's lemma (3/3)

- We only want to collect the terms of order $O(dt)$ or lower, since higher-order terms approach faster to zero. So the terms $dt\,dt$ and $dX_t dt$ drop out, and

$$dX_t dX_t = \sigma_t^2 dW_t dW_t + o(dt) = \sigma_t^2 dt.$$

- We end up with

$$dY_t = \left(\frac{\partial g}{\partial t} + \mu_t \frac{\partial g}{\partial X_t} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 g}{\partial X_t^2} \right) dt + \sigma_t \frac{\partial g}{\partial X_t} dW_t.$$

Itô's lemma: an example

- Consider $g(t, S_t) = \ln S_t$ and

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

- By Itô's lemma,

$$\begin{aligned} dg &= \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 g}{\partial S_t^2} dS_t dS_t = 0 \cdot dt + \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \sigma^2 S_t^2 dt \\ &= \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) - \frac{\sigma^2}{2} dt = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t. \end{aligned}$$

- So

$$\ln S_t = \ln S_{t_0} + \left(\mu - \frac{\sigma^2}{2} \right) (t - t_0) + \sigma (W_t - W_{t_0}).$$

Distribution and density: Normal

- A normal random variable $Z \sim \mathcal{N}(m, s^2)$ with expectation m and standard deviation s has the probability density function (pdf)

$$f_Z(z) = \frac{1}{\sqrt{2\pi}s^2} \exp\left(-\frac{(z-m)^2}{2s^2}\right).$$

- From the GBM solution, we have

$$X = \ln S_t \sim \mathcal{N}\left(\ln S_{t_0} + \left(\mu - \frac{1}{2}\sigma^2\right)(t - t_0), \sigma^2(t - t_0)\right).$$

- Therefore, the pdf of X is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi(t-t_0)}} \exp\left(-\frac{(x - \ln S_{t_0} - (\mu - \frac{1}{2}\sigma^2)(t - t_0))^2}{2\sigma^2(t - t_0)}\right).$$

- The cumulative distribution function (CDF) of X can be written as

$$F_X(x) = \Phi\left(\frac{x - \ln S_{t_0} - (\mu - \frac{1}{2}\sigma^2)(t - t_0)}{\sigma\sqrt{t - t_0}}\right),$$

Distribution and density: Log-normal

- For the log-normal random variable $S = \exp(X)$, its CDF is

$$F_S(y) = \mathbb{P}(S \leq y) = \begin{cases} 0, & y \leq 0, \\ \mathbb{P}(X \leq \ln y), & y > 0. \end{cases}$$

- For $y > 0$,

$$F_S(y) = \int_{-\infty}^{\ln y} f_X(x) dx.$$

- Equivalently, using the log-normal density,

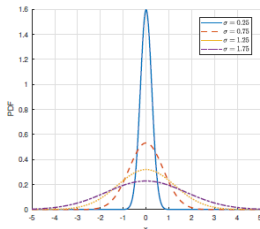
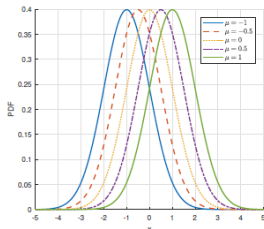
$$F_S(y) = \int_0^y f_S(s) ds, \quad y > 0.$$

- The log-normal pdf is

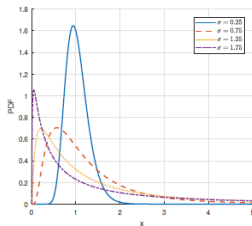
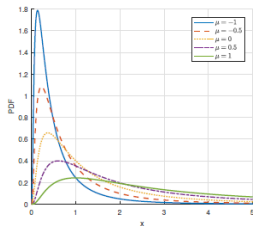
$$f_S(s) = \frac{1}{s\sigma\sqrt{2\pi(t-t_0)}} \exp\left(-\frac{\left(\ln \frac{s}{S_{t_0}} - \left(\mu - \frac{1}{2}\sigma^2\right)(t-t_0)\right)^2}{2\sigma^2(t-t_0)}\right), \quad s > 0.$$

Simulated dynamics and densities (1/2)

- Normal:



- Log-normal:



Simulated dynamics and densities (2/2)

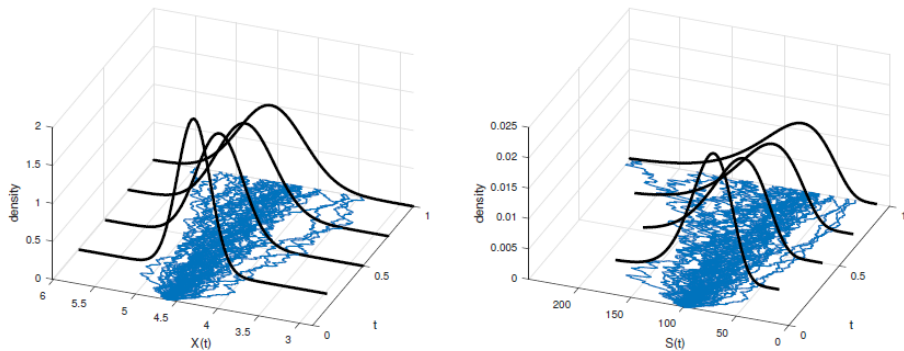


Figure: Paths and the corresponding densities. Left: $X(t) = \log S(t)$ and Right: $S(t)$ with the following configuration: $S_0 = 100$, $\mu = 0.05$, $\sigma = 0.4$; $T = 1$.

Black–Scholes PDE for European options (1/4)

- Assume the following stochastic dynamics for the underlying asset price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t^P,$$

and for the bank account:

$$dM_t = rM_t dt,$$

where the superscript P in W_t^P denotes the real-world (physical) measure.

- Let $V(t, S_t)$ denote the option price at time t .
 - Consider a continuous, self-financing hedging strategy that is long one unit of the option and short $\Delta(t, S_t)$ units of the underlying asset. Self-financing means that no additional capital is injected into or withdrawn from the portfolio.
- The value of the hedged portfolio is therefore

$$\Pi(t, S_t) = V(t, S_t) - \Delta(t, S_t)S_t.$$

Black–Scholes PDE for European options (2/4)

- Since the strategy is self-financing, the change in the hedged portfolio value over $[t, t + dt]$ is

$$d\Pi_t = dV_t - \Delta(t, S_t) dS_t.$$

- The position $\Delta(t, S_t)$ is chosen based on information available at time t and therefore remains fixed over $[t, t + dt]$.
- To compute dV_t , apply Itô's lemma to $V(t, S_t)$:

$$dV_t = V_t dt + V_S dS_t + \frac{1}{2} V_{SS} (dS_t)^2.$$

- Using

$$dS_t = \mu S_t dt + \sigma S_t dW_t^P, \quad (dS_t)^2 = \sigma^2 S_t^2 dt,$$

we obtain

$$dV_t = \left(V_t + \mu S_t V_S + \frac{1}{2} \sigma^2 S_t^2 V_{SS} \right) dt + \sigma S_t V_S dW_t^P.$$

Black–Scholes PDE for European options (3/4)

- Substituting dV_t and dS_t into $d\Pi_t$ gives

$$d\Pi_t = \left(V_t + \mu S_t V_S + \frac{1}{2} \sigma^2 S_t^2 V_{SS} - \Delta(t, S_t) \mu S_t \right) dt \\ + (\sigma S_t V_S - \Delta(t, S_t) \sigma S_t) dW_t^P.$$

- The portfolio is locally risk-free if and only if the Brownian motion term vanishes, i.e.,

$$V_S - \Delta(t, S_t) = 0.$$

- Choosing

$$\Delta(t, S_t) = V_S,$$

we obtain

$$d\Pi_t = \left(V_t + \frac{1}{2} \sigma^2 S_t^2 V_{SS} \right) dt.$$

Black–Scholes PDE for European options (4/4)

- Since the hedged portfolio is risk-free, the no-arbitrage principle implies that it must earn the risk-free rate:

$$d\Pi_t = r\Pi_t dt.$$

- Using

$$\Pi_t = V(t, S_t) - V_S S_t,$$

we obtain

$$V_t + \frac{1}{2}\sigma^2 S_t^2 V_{SS} = r(V - S_t V_S).$$

- Rearranging yields the Black–Scholes PDE:

$$V_t + rS V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0.$$

- The Black–Scholes PDE is a second-order linear parabolic partial differential equation.

Main assumptions of Black–Scholes

- The asset price follows the diffusion process

$$dS_t = \mu S_t dt + \sigma S_t dW_t^P.$$

- Interest rate r and volatility σ are known constants.
- Transaction costs for hedging are not included in the model.
- No dividend is paid during the life of the option.
- There are no arbitrage possibilities.

- Greeks are sensitivities of the option price w.r.t. model inputs.
- **Delta:**

$$\Delta = \frac{\partial V}{\partial S_t}$$

is the rate of change of the value of the option w.r.t. the price of the underlying asset S_t . It is the unit of the underlying asset required to delta-hedge the option in order to eliminate the risk due to fluctuations of S_t .

- **Gamma:** indicates the change in Delta

$$\Gamma = \frac{\partial^2 V}{\partial S^2}.$$

If Gamma is small, Delta is stable and thus in practice it is only necessary to delta-hedge from time to time. If Gamma is high, the hedging position has to be adjusted frequently.

- There are several other important hedging parameters, e.g., *theta*, which is the sensitivity of the option value to the time to maturity.

What determines the value of an option

- The value of a call option is a function of parameters in the contract and the properties of the underlying asset, including
 - The strike price K .
 - The time to maturity $T - t$. T is the expiry time and t is the current time.
 - The current asset price S_t .
 - The volatility of the asset return σ .
 - The interest rate r .
- The famous Black–Scholes formula of the European call option, which is a solution to the Black–Scholes PDE, is a function of the 5 parameters listed above, i.e.,

$$V(S, t, \sigma, K, T, r).$$

- Note that the parameter μ in the Black–Scholes model, i.e., the growth rate of the asset price, does not appear in the Black–Scholes PDE and hence also not in the Black–Scholes formula of the option price.

- **Definition (Martingale):** A stochastic process $X = \{X_t; t \geq 0\}$ is a martingale with respect to a filtration $\{\mathcal{F}_t\}$ if:
 - X_t is adapted to $\{\mathcal{F}_t\}$,
 - $\mathbb{E}(|X_t|) < \infty$ for all $t \geq 0$,
 - for all $0 \leq s < t$,

$$\mathbb{E}(X_t \mid \mathcal{F}_s) = X_s.$$

- **Exercise:** Show that Brownian motion W_t is a martingale.
 - W_t is adapted to its natural filtration.
 - $\mathbb{E}(|W_t|) < \infty$ since $W_t \sim N(0, t)$.
 - For $s < t$,

$$\mathbb{E}(W_t \mid \mathcal{F}_s) = \mathbb{E}(W_t - W_s \mid \mathcal{F}_s) + W_s.$$

Since $W_t - W_s$ is independent of $\mathcal{F}_s$ and $\mathbb{E}(W_t - W_s) = 0$, we obtain

$$\mathbb{E}(W_t \mid \mathcal{F}_s) = W_s.$$

Martingale: examples (1/2)

- Let W_t be a Brownian motion on $(\Omega, \mathcal{F}, P)$. Define

$$X_t = \exp\left(\sigma W_t - \frac{1}{2}\sigma^2 t\right).$$

Then $\{X_t\}$ is a martingale.

- We use the Gaussian exponential moment: $\mathbb{E}\left[e^{\sigma Z - \frac{1}{2}\sigma^2}\right] = 1, \quad Z \sim N(0, 1)$.
- To verify the martingale property:
 - X_t is adapted since it is a function of W_t .
 - $\mathbb{E}(|X_t|) = 1$.
 - For $s < t$,

$$\mathbb{E}(X_t \mid \mathcal{F}_s) = \mathbb{E}\left(e^{\sigma W_t - \frac{1}{2}\sigma^2 t} \mid \mathcal{F}_s\right) = e^{\sigma W_s - \frac{1}{2}\sigma^2 s} \mathbb{E}\left(e^{\sigma(W_t - W_s) - \frac{1}{2}\sigma^2(t-s)}\right).$$

Since $W_t - W_s \sim N(0, t - s)$, $\mathbb{E}\left(e^{\sigma(W_t - W_s) - \frac{1}{2}\sigma^2(t-s)}\right) = 1$, it holds that

$$\mathbb{E}(X_t \mid \mathcal{F}_s) = X_s.$$

Martingale: examples (2/2)

- Let $g(t)$ be an adapted process satisfying

$$\mathbb{E} \left[\int_0^T g^2(s) ds \right] < \infty.$$

- Then the stochastic integral

$$X_t = \int_0^t g(s) dW_s, \quad t \in [0, T],$$

is a martingale with respect to $\{\mathcal{F}_t\}$.

- In particular,

$$\mathbb{E}(X_t \mid \mathcal{F}_s) = X_s, \quad 0 \leq s < t \leq T.$$

Martingale approach for option pricing (1/3)

- In mathematical finance, a *risk-neutral measure* Q (also called an *equivalent martingale measure*) is a probability measure:
 - equivalent to the real-world measure P ;
 - under which the discounted asset price is a martingale.

- That is,

$e^{-rt}S_t$ is a martingale under Q .

- Equivalently,

$$S_t = \mathbb{E}^Q \left[e^{-r(T-t)} S_T \mid \mathcal{F}_t \right].$$

- Under Q , suppose

$$dS_t = \mu' S_t dt + \sigma S_t dW_t^Q.$$

If $e^{-rt}S_t$ is a martingale, what is μ' ?

Martingale approach for option pricing (2/3)

- Apply Itô's product rule:

$$d(e^{-rt}S_t) = S_t d(e^{-rt}) + e^{-rt}dS_t + d(e^{-rt})dS_t \quad (1)$$

$$= -re^{-rt}S_tdt + e^{-rt}(\mu'S_tdt + \sigma S_t dW_t^Q) \quad (2)$$

$$= (\mu' - r)e^{-rt}S_tdt + e^{-rt}\sigma S_t dW_t^Q. \quad (3)$$

- Integrating,

$$e^{-rt}S_t = S_0 + \int_0^t (\mu' - r)e^{-rv}S_vdv + \int_0^t e^{-rv}\sigma S_v dW_v^Q.$$

- The stochastic integral is a martingale. Therefore, $e^{-rt}S_t$ is a martingale if and only if the drift term vanishes, i.e.

$$\mu' - r = 0.$$

- Hence under Q ,

$$dS_t = rS_tdt + \sigma S_t dW_t^Q.$$

Martingale approach for option pricing (3/3)

- Under no-arbitrage in the Black–Scholes model, the discounted option price is a martingale under Q :

$$e^{-rt}V(t, S_t) \text{ is a martingale.}$$

- Using Itô's lemma and the product rule,

$$d(e^{-rt}V) = -re^{-rt}Vdt + e^{-rt}dV.$$

- Under Q , since

$$dS_t = rS_tdt + \sigma S_t dW_t^Q,$$

Itô's lemma gives

$$dV = \left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma SV_S dW_t^Q.$$

- Therefore,

$$d(e^{-rt}V) = e^{-rt} \left(V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV \right) dt + e^{-rt}\sigma SV_S dW_t^Q.$$

Connecting the PDE and martingale approaches

- We have derived the Black–Scholes equation using:
 - the replication (PDE) approach;
 - the martingale approach under a risk-neutral measure.
- These two approaches lead to the same pricing equation.
- The Feynman–Kac formula provides the mathematical link between parabolic PDEs and conditional expectations, thereby connecting the two approaches.

Feynman–Kac formula (1/3)

- The Feynman–Kac formula establishes a link between parabolic PDEs and diffusion processes.
- Consider the PDE

$$\frac{\partial V}{\partial t} + b(x, t) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2(x, t) \frac{\partial^2 V}{\partial x^2} - r(x, t) V = 0,$$

with terminal condition

$$V(x, T) = g(x).$$

Feynman–Kac formula (2/3)

- Let X_s solve the diffusion

$$dX_s = b(X_s, s) ds + \sigma(X_s, s) dW_s, \quad s \geq t,$$

with initial condition $X_t = x$.

- Then the Feynman–Kac theorem states:

$$V(x, t) = \mathbb{E} \left[e^{-\int_t^T r(X_u, u) du} g(X_T) \mid X_t = x \right].$$

- Thus the solution to the PDE is represented as a discounted conditional expectation.

Feynman–Kac formula (3/3)

- Define the process

$$Y_s = e^{-\int_t^s r(X_u, u) du} V(X_s, s), \quad s \geq t.$$

- Applying Itô's lemma,

$$\begin{aligned} dY_s = & e^{-\int_t^s r(X_u, u) du} \left(V_t + b(X_s, s) V_x + \frac{1}{2} \sigma^2(X_s, s) V_{xx} - r(X_s, s) V \right) ds \\ & + e^{-\int_t^s r(X_u, u) du} \sigma(X_s, s) V_x dW_s. \end{aligned}$$

- By the PDE condition, the drift term vanishes. Hence

$$dY_s = e^{-\int_t^s r(X_u, u) du} \sigma(X_s, s) V_x dW_s.$$

- Therefore Y_s is a martingale, and

$$V(x, t) = \mathbb{E} \left[e^{-\int_t^T r(X_u, u) du} g(X_T) \mid X_t = x \right].$$

Feynman–Kac: an example

- **Exercise:** Solve the PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial x^2} = 0, \quad V(x, T) = x^2,$$

where σ is constant.

- **Solution:** By Feynman–Kac,

$$V(x, t) = \mathbb{E} [X_T^2 \mid X_t = x],$$

where

$$dX_s = \sigma dW_s, \quad X_t = x.$$

- Hence

$$X_T = x + \sigma(W_T - W_t) \sim \mathcal{N}(x, \sigma^2(T - t)).$$

- Using

$$\mathbb{E}(X_T^2) = \text{Var}(X_T) + (\mathbb{E}X_T)^2,$$

we obtain

$$V(x, t) = \sigma^2(T - t) + x^2.$$

Feynman–Kac: connecting the PDE and martingale approaches

- For the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0,$$

with terminal condition

$$V(S, T) = g(S),$$

- Feynman–Kac implies

$$V(S_t, t) = e^{-r(T-t)} \mathbb{E}^Q [g(S_T) \mid \mathcal{F}_t],$$

where under Q

$$dS_t = rS_t dt + \sigma S_t dW_t^Q.$$

- Rewriting,

$$e^{-rt} V(S_t, t) = \mathbb{E}^Q [e^{-rT} g(S_T) \mid \mathcal{F}_t].$$

- Thus the discounted price

$$\frac{V_t}{e^{rt}}$$

is a martingale. The process $B_t = e^{rt}$ is called the numeraire.

Back to risk-neutral measure

- The risk-neutral measure Q in the Black–Scholes model has the following properties:
 - Q is equivalent to the real-world measure P (they assign zero probability to the same events).
 - Under Q , the asset price satisfies

$$dS_t = rS_t dt + \sigma S_t dW_t^Q.$$

The drift changes from μ to r , while the diffusion coefficient σ remains unchanged.

- Pricing depends only on parameters under Q and the payoff function. The real-world drift μ does not affect arbitrage-free prices.
- In practice, one can either:
 - derive the PDE via replication, or
 - express the price as a risk-neutral expectation and recover the PDE using Feynman–Kac.

Big picture: replication, PDE, and martingale pricing

Model setup (Black–Scholes under P): $dS_t = \mu S_t dt + \sigma S_t dW_t^P$, $B_t = e^{rt}$

↓ **No-arbitrage + completeness** ↓

Replication / hedging (PDE approach)

Choose Δ_t to eliminate risk locally.

⇒ **Black–Scholes PDE:**

$$V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0, \quad V(S, T) = g(S).$$

Change of measure (martingale approach)

Exist $Q \sim P$ such that discounted prices are martingales.

Under Q :

$$dS_t = rS_t dt + \sigma S_t dW_t^Q.$$

↓ **Feynman–Kac links PDE ↔ expectation under Q** ↓

Risk-neutral valuation (pricing formula):

$$V(S_t, t) = \mathbb{E}^Q \left[e^{-r(T-t)} g(S_T) \mid \mathcal{F}_t \right].$$

Equivalently, the discounted option price is a martingale:

Equivalent Martingale Measure (EMM)

- Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$ be a filtered probability space, and let S_t denote the asset price process. Let B_t be the bank account (numeraire).
- A probability measure Q is called an **Equivalent Martingale Measure (EMM)** if:
 - **(Equivalence)** Q is equivalent to P , i.e.

$$Q(A) = 0 \iff P(A) = 0.$$

The two measures agree on which events are impossible.

- **(Martingale property)** The discounted asset price is a martingale under Q :

$$\frac{S_t}{B_t} = \mathbb{E}^Q \left[\frac{S_T}{B_T} \middle| \mathcal{F}_t \right].$$

- In the Black–Scholes model,

$$dS_t = rS_t dt + \sigma S_t dW_t^Q,$$

so prices can be computed as

$$V_t = B_t \mathbb{E}^Q \left[\frac{g(S_T)}{B_T} \middle| \mathcal{F}_t \right].$$

Computational methods at our disposal

- We now have several computational approaches for pricing:
 - **Solve the PDE** (analytically or numerically).
 - **Compute the risk-neutral expectation**

$$V(S_t, t) = e^{-r(T-t)} \mathbb{E}^Q [g(S_T) \mid \mathcal{F}_t].$$

- If the risk-neutral density exists,

$$V(S_t, t) = e^{-r(T-t)} \int g(s) f_{S_T|S_t}(s \mid S_t) ds.$$

- $f_{S_T|S_t}$ is the conditional risk-neutral density of S_T .
- In many models, the characteristic function is analytically or semi-analytically available.
- Monte Carlo simulation.
- Numerical integration.
- Fourier methods.

Analytical solutions of the Black–Scholes PDE (1/5)

- We derive the classical Black–Scholes formula for a European call option.
- Under the risk-neutral measure Q ,

$$dS_t = rS_t dt + \sigma S_t dW_t^Q.$$

- Apply Itô's lemma to $\ln S_t$:

$$\begin{aligned} d \ln S_t &= \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} (dS_t)^2 \\ &= \left(r - \frac{\sigma^2}{2} \right) dt + \sigma dW_t^Q. \end{aligned}$$

- Hence,

$$\ln S_T = \ln S_0 + \left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T^Q.$$

- Therefore,

$$S_T = S_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) T + \sigma W_T^Q \right).$$

Analytical solutions of the Black–Scholes PDE (2/5)

- The time-0 price of a European call with strike K and maturity T is

$$C_0 = e^{-rT} \mathbb{E}^Q[(S_T - K)^+].$$

- Since $S_T = S_0 e^Z$ with

$$Z \sim \mathcal{N}\left(\left(r - \frac{\sigma^2}{2}\right) T, \sigma^2 T\right),$$

we rewrite

$$C_0 = e^{-rT} \int_{\ln(K/S_0)}^{\infty} (S_0 e^z - K) f_Z(z) dz.$$

Analytical solutions of the Black–Scholes (3/5)

$$\int_{\ln(K/S_0)}^{\infty} K f_Z(z) dz = K \mathbb{P}\left(Z \geq \ln \frac{K}{S_0}\right).$$

Standardizing:

$$\begin{aligned} &= K \mathbb{P}\left(N(0, 1) \geq \frac{\ln(K/S_0) - \left(r - \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}\right) \\ &= K \mathbb{P}(N(0, 1) \leq d_2) = K N(d_2), \end{aligned}$$

where

$$d_2 = \frac{\ln(S_0/K) + \left(r - \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}.$$

Analytical solutions of the Black–Scholes (4/5)

$$\int_{\ln(K/S_0)}^{\infty} S_0 e^z f_Z(z) dz = S_0 e^{rT} \mathbb{P}(N(0,1) \leq d_1) = S_0 e^{rT} N(d_1),$$

where

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}.$$

Analytical solutions of the Black–Scholes (5/5)

- Combining the two parts,

$$C_0 = S_0 N(d_1) - Ke^{-rT} N(d_2).$$

- This is the Black–Scholes formula for a European call option.
- The European put price follows either by a similar derivation or from put–call parity:

$$P_0 = C_0 - S_0 + Ke^{-rT}.$$